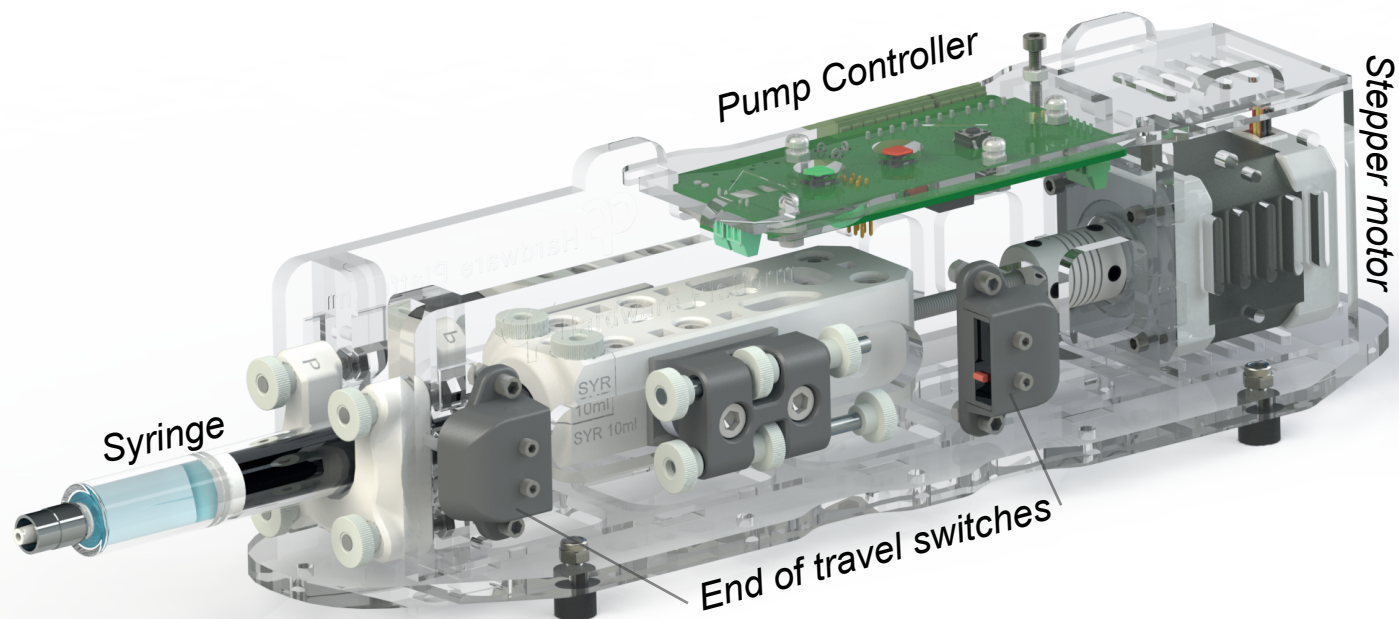


a

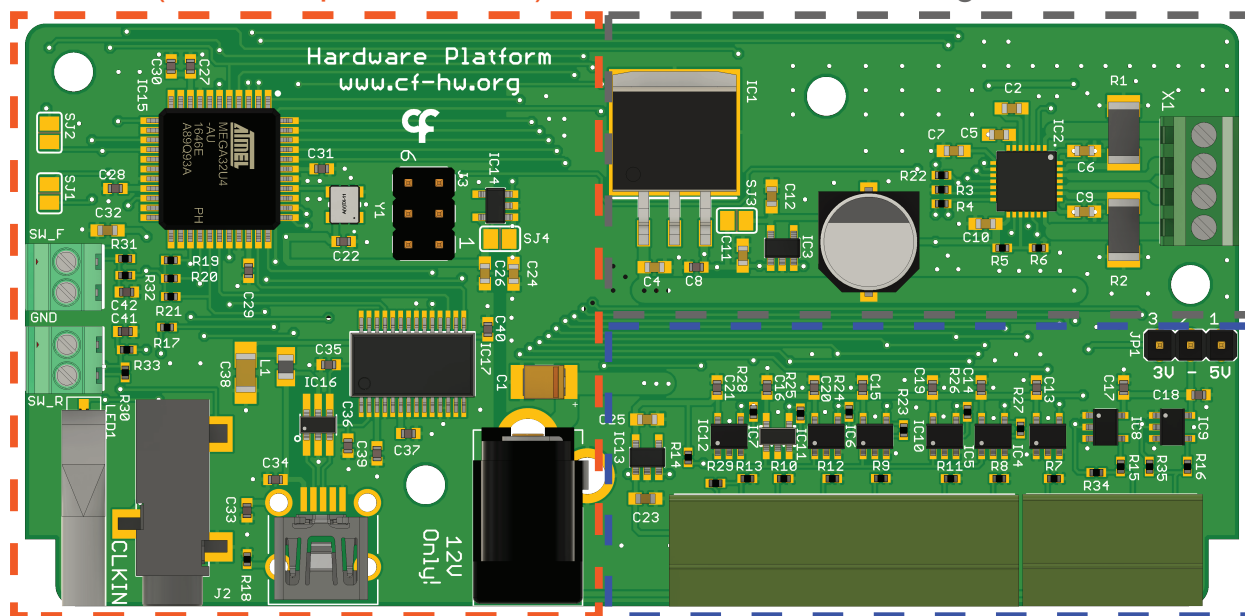


b

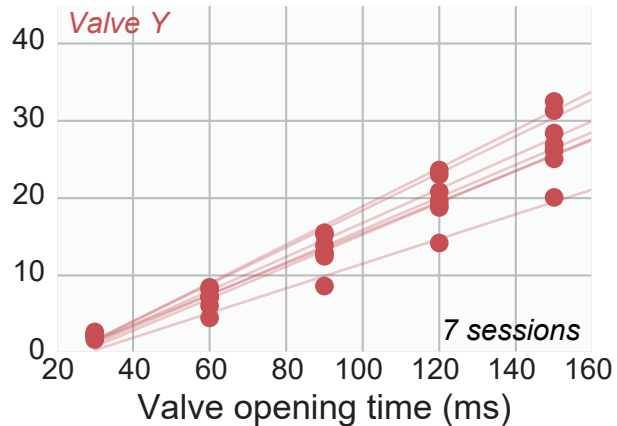
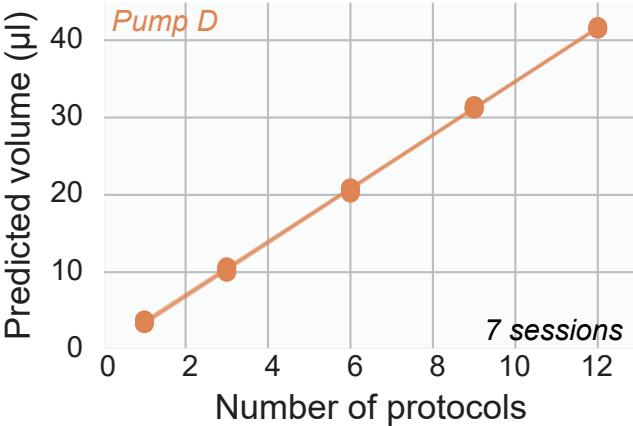
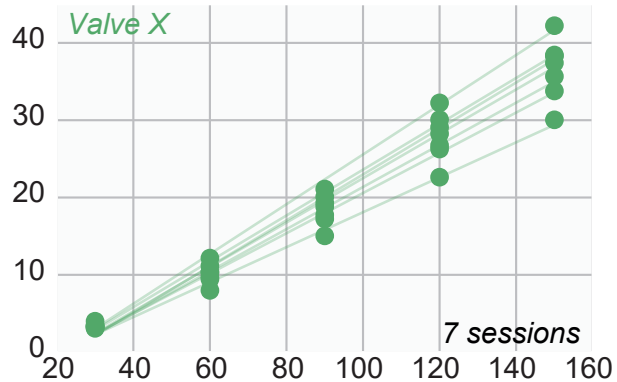
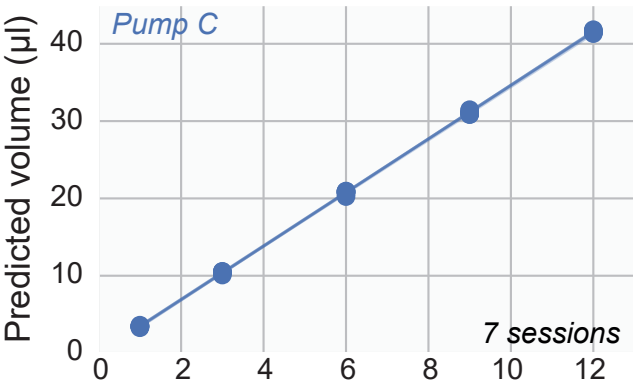
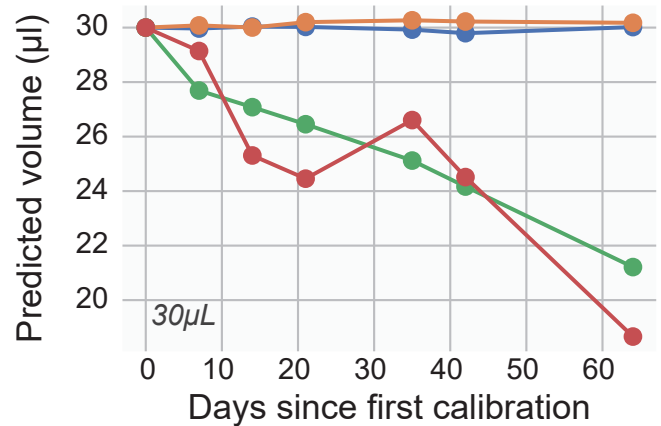
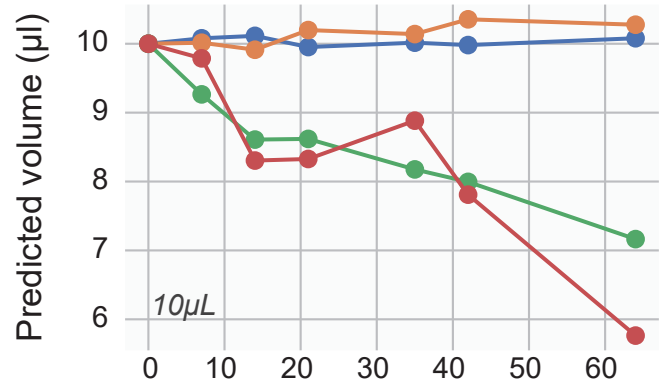
Pump Controller PCB

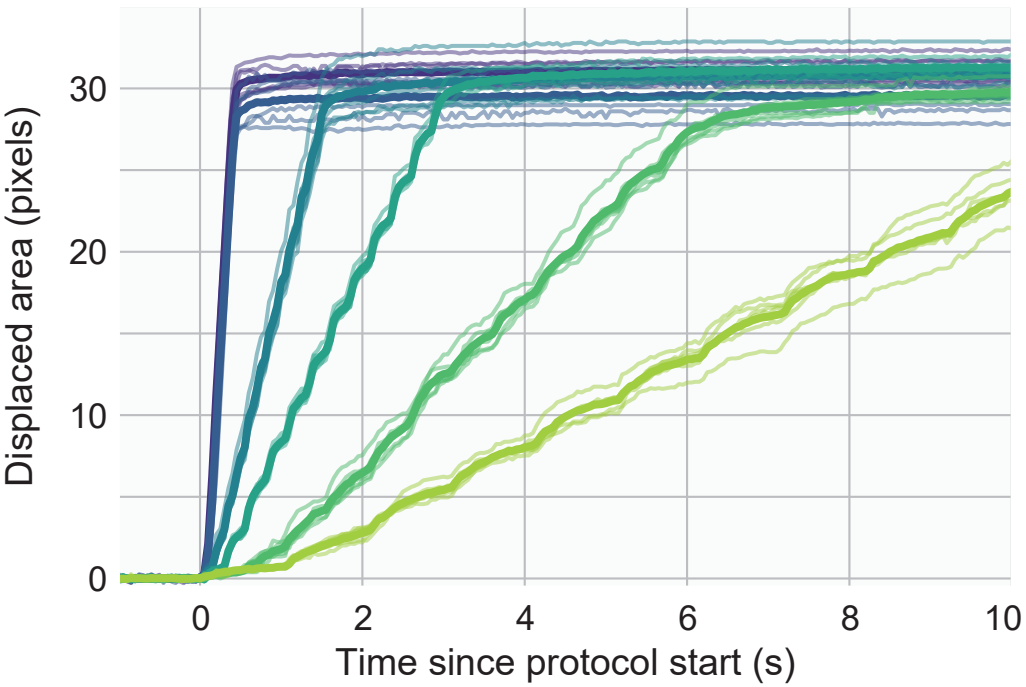
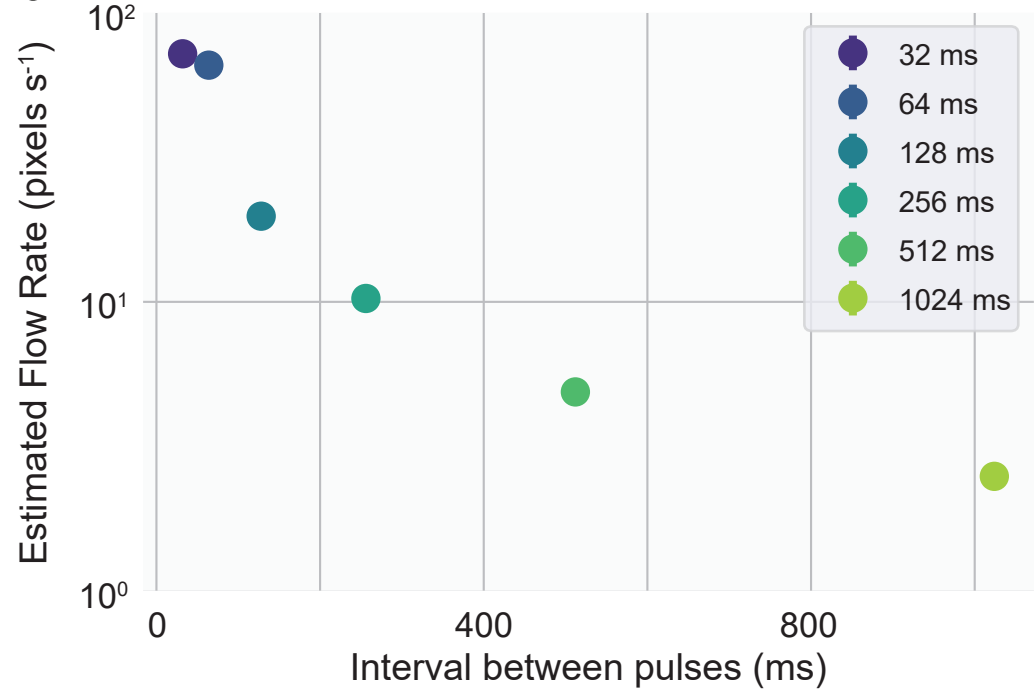
Microcontroller
(HARP implementation)

Stepper motor driver
Power management



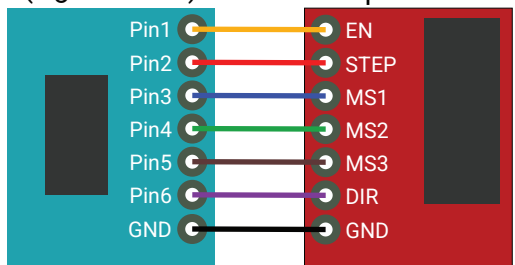
IO breakout

a**b**

a**b**

a

Low-level hardware control

Microcontroller
(e.g. Arduino)

ControlPump.ino

```
(...)

void loop() {

    //Pulse with a 4s period
    //by toggling STEP input

    digitalWrite(Pin2, HIGH);
    delay(2000);
    digitalWrite(Pin2, LOW);
    delay(2000);
}
```

b

Hybrid control

Configuration

TTL Source



Pump controller

Events

- ☒ Step state
- ☒ Direction state
- ☒ Switch forward state
- ☒ Switch reverse state
- ☒ Input state
- ☒ Protocol state

Protocol

Protocol direction

Reverse

Protocol type

Step

Number of steps

15

Step period [ms]

10

Input/Output

Digital input

Use as a pure digital input

Digital output 0

Digital output controlled by software

Digital output 1

Digital output controlled by software

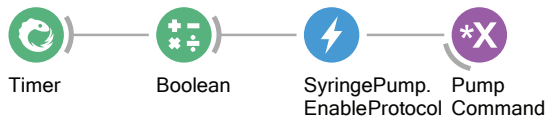
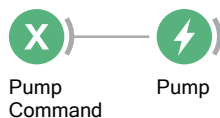
Other

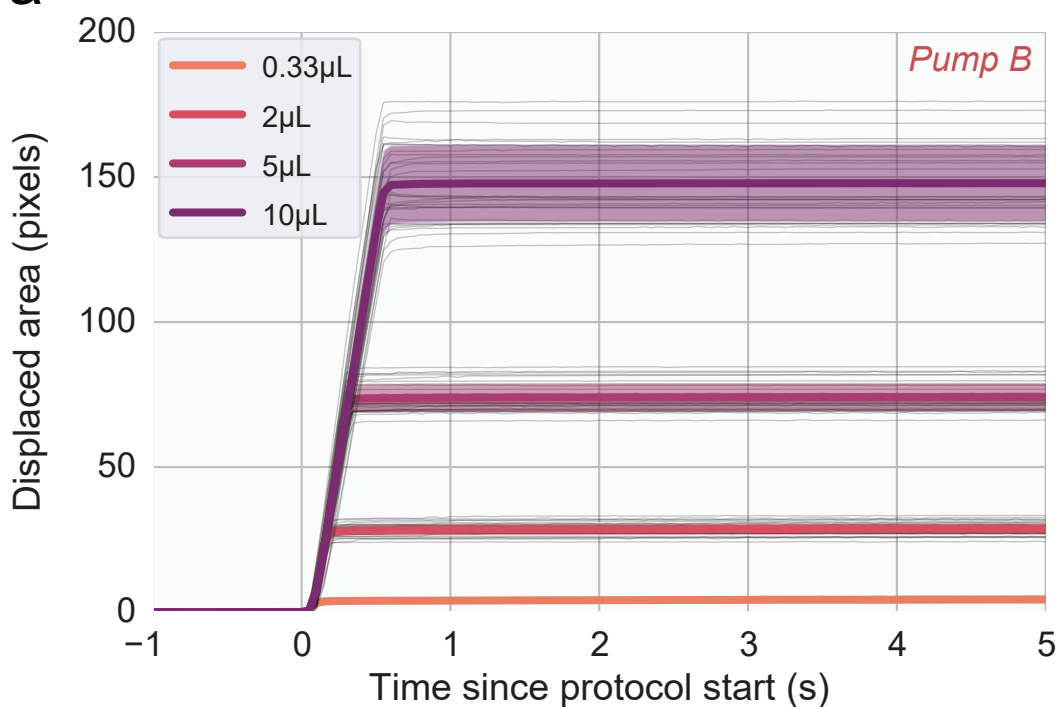
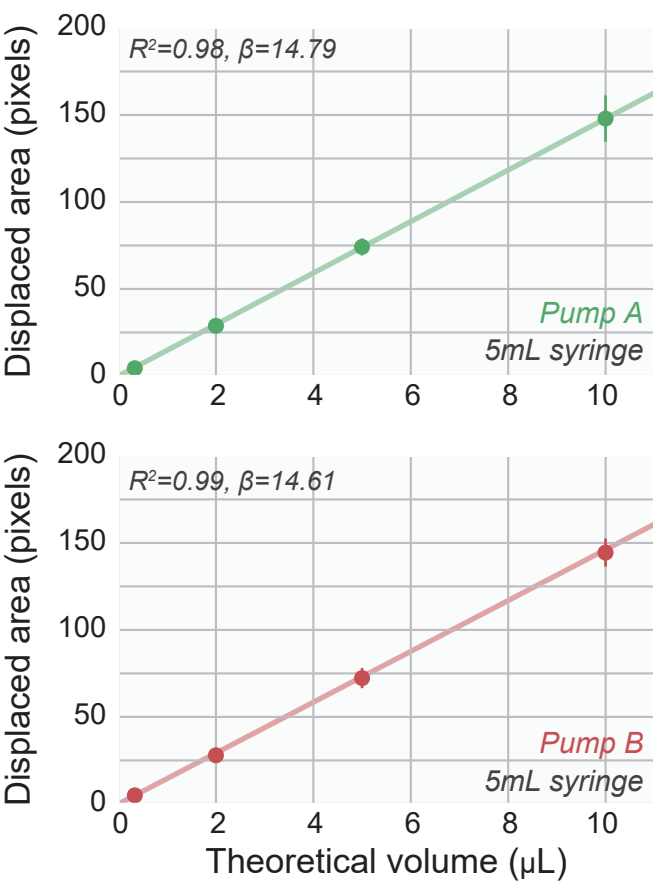
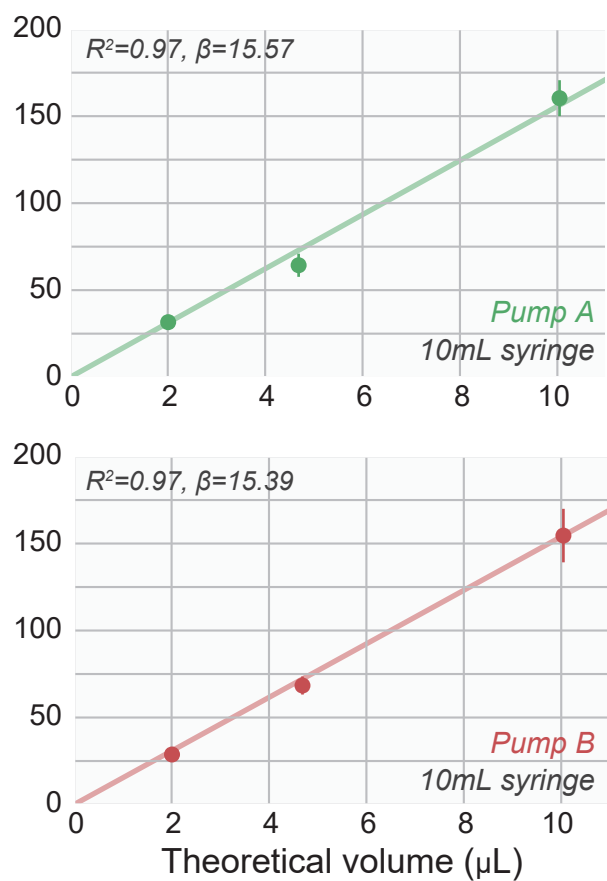
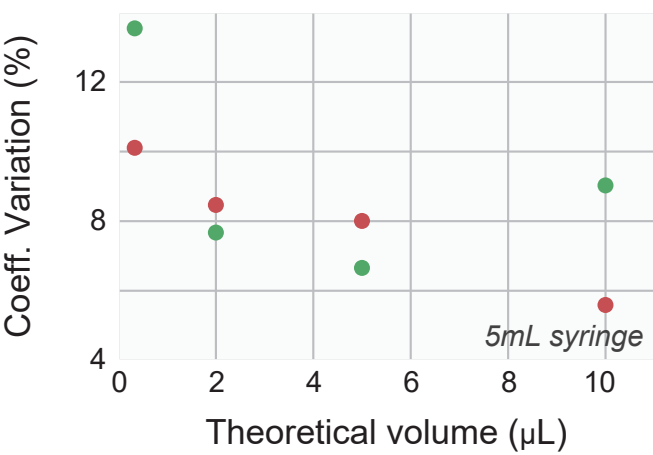
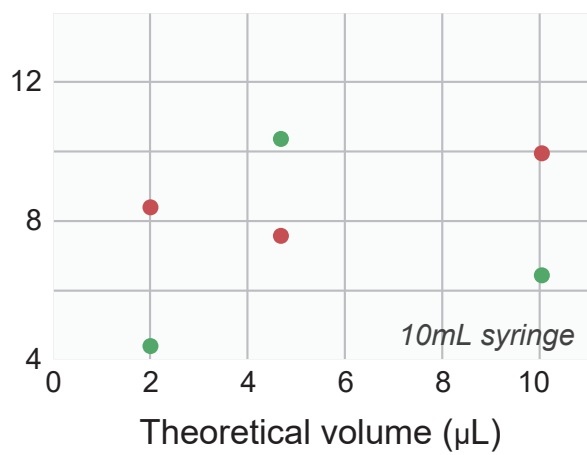
Motor microstep

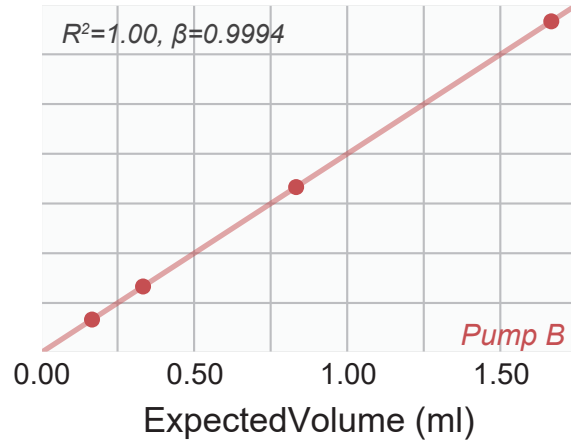
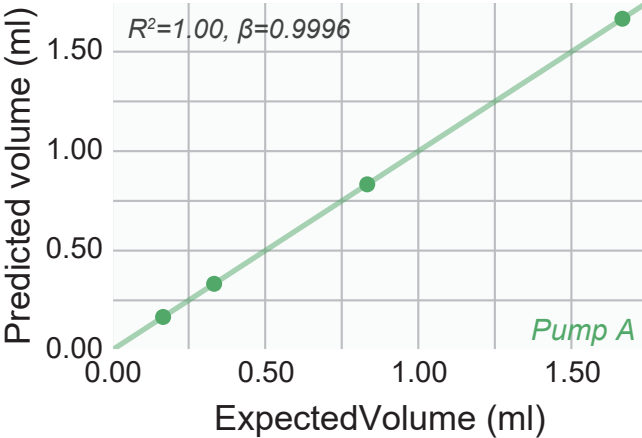
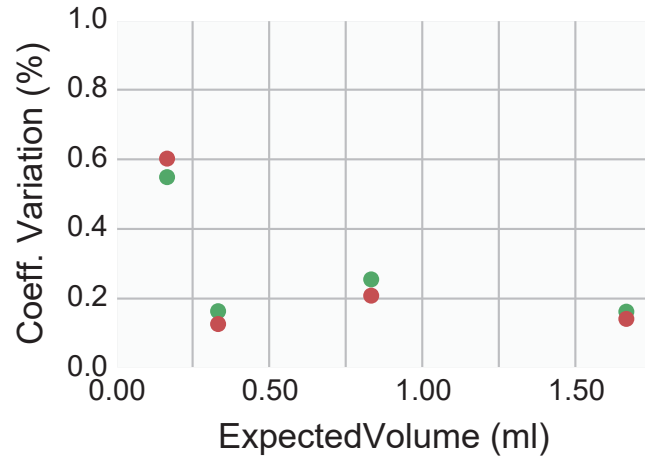
Full step

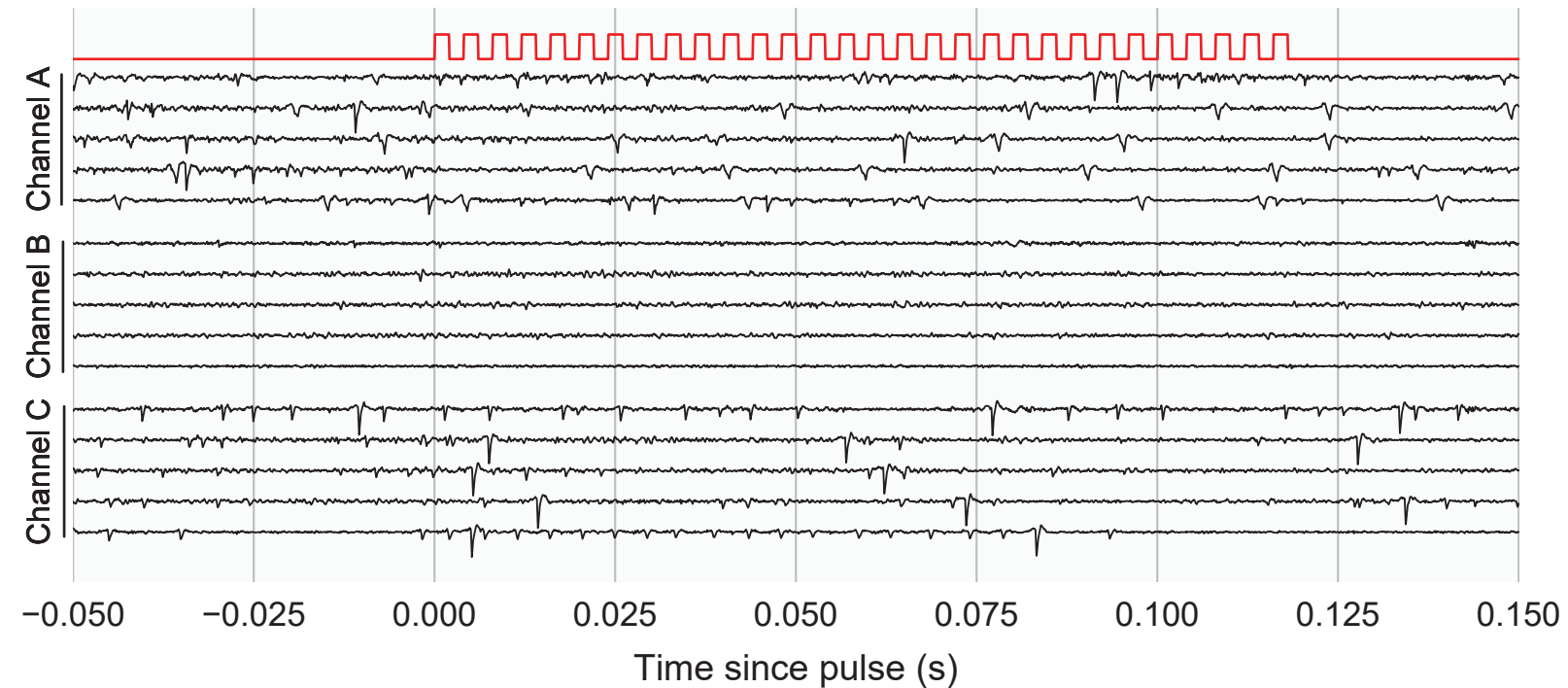
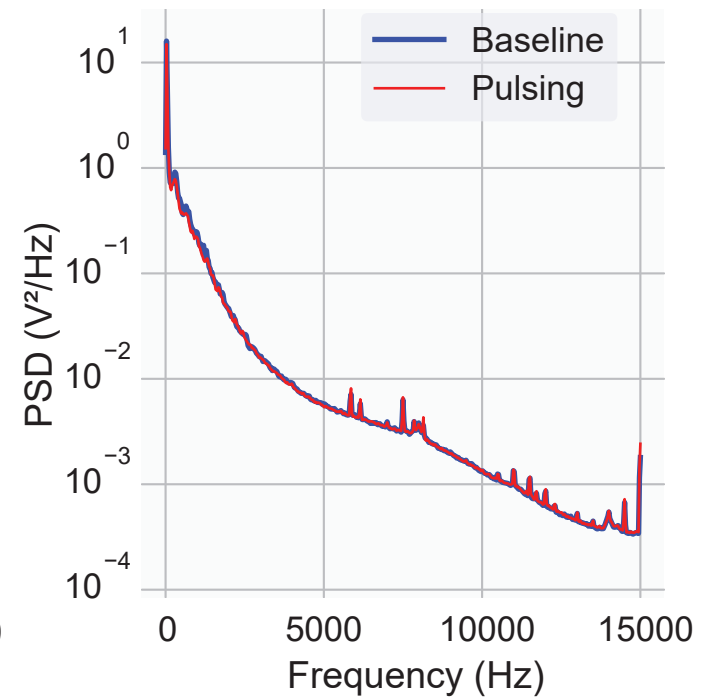
c

Software control

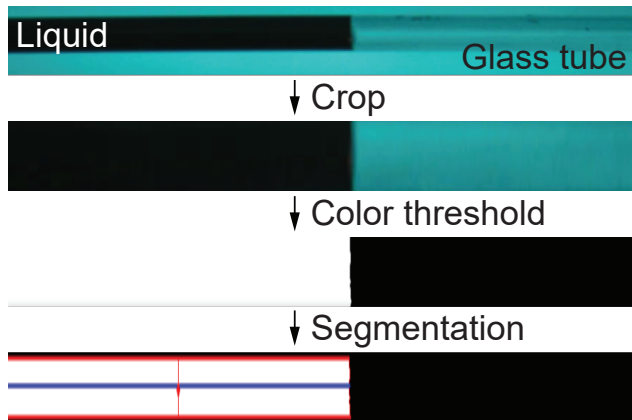
bonsai
VISUAL REACTIVE PROGRAMMING

a**b****c****d****e**

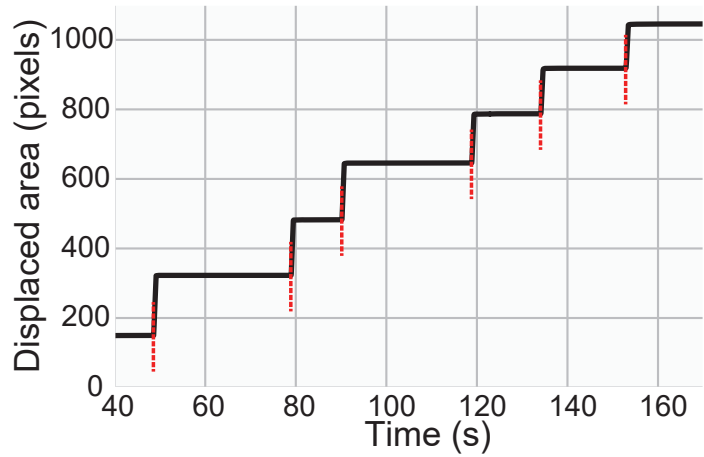
a**b**

a**b**

a

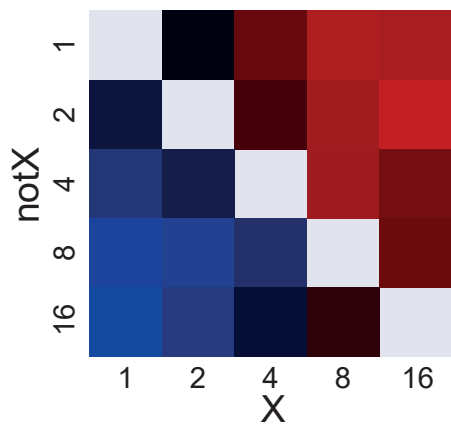


b

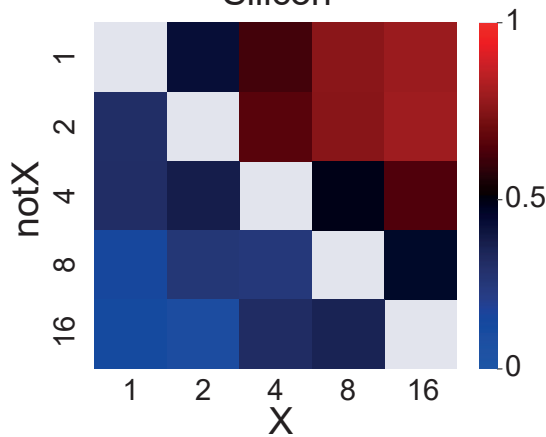


a

Aluminium



Silicon

**b**